

Chaitanya Tripathi
Ashoka University, Sonipat, Haryana 131029, India ORCID 0009-0002-4094-0323
chaitanya.tripathi_ug2025@ashoka.edu.in

3 October 2026

Professor Olga Korotkova
Editor-in-Chief, *Journal of the Optical Society of America A*

Dear Professor Korotkova,

I am submitting the manuscript “Media-in-the-loop holography: designing photopolymer exposures through a differentiable recording model” for consideration as a Research Article in *JOSA A*.

Camera-in-the-loop correction has substantially improved computer-generated holography on rewritable modulators, but it cannot be used with write-once volume photopolymers, where the reconstruction error can be measured only after it is permanent. The manuscript studies, in simulation, the alternative of designing the recording exposure through a differentiable model of the recording itself: the non-local polymerization-driven diffusion (NPDD) model introduced in this journal by Sheridan and Lawrence [*JOSA A* **17**, 1108 (2000)], coupled to split-step beam-propagation readout.

On a benchmark of 30 spatial-frequency and contrast-budget cells, designing through the model improves on the same optimizer with a linear, media-blind recording assumption in 29 cells (by up to 2.52 dB) and reduces the fraction of cells more than 2 dB short of an oracle reference from 37% to 0%. A factorial ablation with a slope-matched linear control shows that the gain requires a nonlinear dose response but no single saturating mechanism. The study also reports where the approach fails: the gain decays toward zero at high spatial frequency, largely vanishes for fringes slanted by 10° or more, and reverses if the model underestimates the medium’s index budget several-fold. The recording model reproduces published Bayfol HX dose-response data, including a parameter-free prediction between a measured and a modelled series.

The work sits within *JOSA A*’s core topics of holography, diffraction gratings and computational imaging, and should interest readers working on holographic recording media, computer-generated holography and physics-based inverse design. The claims are scoped to what the simulations support: every number in the manuscript is generated by script from released per-job results, and the code and data are public (Data availability).

Declarations. The manuscript is original, has not been published, and is not under consideration by another journal. A companion study (Part 1, on phase-only CGH with idealized modulators) is available as a preprint on Optica Open and is cited in the manuscript; it does not overlap with the present results. The author declares no conflicts of interest and received no external funding. Use of a generative AI assistant is disclosed in the Acknowledgment, as Optica policy requires. The manuscript runs to 10 pages in the Optica universal template (about 7 pages in the two-column journal layout), with four figures, three tables and a supplemental document. Suggested reviewers, with contact details, are entered in Prism.

Thank you for considering the manuscript.

Sincerely,

Chaitanya Tripathi